

Reinforcement Learning-Based Optimal Setpoint Determination in Energy Utility Systems

MSc Artificial Intelligence
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Presentation roadmap

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01

Problem & method

Research question
MDP formulation
SAC and TD3

02

Battery case

Storage arbitrage
Terminal SOC
Performance and speed

03

Factory case

Nonlinear steam utility
Reduced policy actions
Constraint limitations

04

Conclusions

Cross-case evidence
Future improvements
Questions

One comparison throughout: economics, feasibility, and online time.

Why setpoint determination matters

- Prices and renewable availability vary over time.
- Charging now changes the energy available later.
- Power, storage, and process limits must be respected.
- Repeated scheduling creates a need for rapid decisions.

LOW-PRICE PERIOD

Charge storage
Buy energy for future use

↓ Stored energy connects the hours

HIGH-PRICE PERIOD

Discharge storage
Reduce purchases or sell energy

Storage couples present costs with future operating flexibility.

Thesis: Sections 1 and 2.1

Research question and success criteria

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Can reinforcement learning determine feasible and cost-effective operating setpoints for energy utility systems of varying complexity?

ECONOMICS

Original grid cost
Battery profit

FEASIBILITY

Physical bounds
Process coupling

ONLINE TIME

Complete schedule
Computation effort

Contributions of the thesis

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01 FINITE-HORIZON FORMULATION

Continuous-action MDPs for battery and factory scheduling.

02 ENGINEERING ACTION MAPS

Map actions into physical setpoints with selected feasibility handling.

03 REDUCED FACTORY ACTIONS

Two policy actions: storage mode and mass flow. Derive other setpoints.

04 MATCHED BENCHMARKS

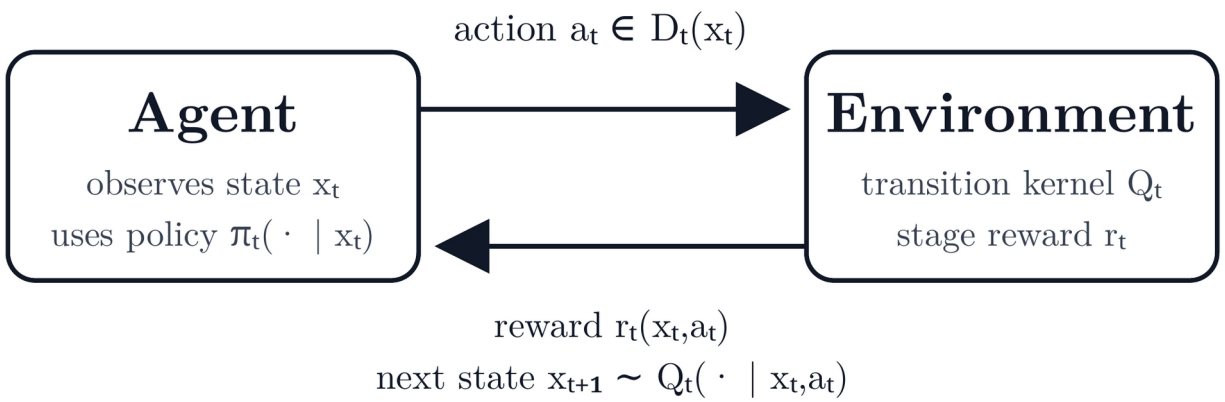
Evaluate cost, feasibility, online time, and offline training.

Established RL algorithms are formulated and evaluated for two energy systems.

Experimental design across two cases

Property	Battery system	Cookie-factory utility
Storage state	State of charge	TES temperature
External inputs	Electricity price	Electricity price + wind
Independent decisions	1 baseline / 1 policy output	5 baseline / 2 policy output
Scheduling horizon	24 hourly decisions	24 hourly decisions
Comparison windows	1000 windows	200 windows

From scheduling problem to MDP



- Observation: storage, inputs, horizon position, forecasts.
- Action: continuous output mapped into physical setpoints.
- Transition: simulator updates storage and process behavior.
- Reward: negative cost plus feasibility penalties.

At each hour: observe flexibility, select an action, and evaluate its consequences.

What makes scheduling a Markov decision process?

Finite-horizon MDP: $(E, A, D_t, Q_t, r_t, g_N)$

Element	Meaning in the energy-system task
State $x_t \in E$	Storage, horizon position, and known price/wind profiles
Action $a_t \in A$; admissible $D_t(x_t)$	Battery command u; factory bypass command b and HTF flow
Transition kernel Q_t	Simulator determines the next storage/process state
Stage reward r_t	Negative grid cost and penalties for remaining violations
Terminal reward	Zero here: terminal penalties are in the final stage reward

Markov: current state + action contain the information needed to predict the next state.

The state summarizes the information needed for the next decision and transition.

Return, state value, and action value

$$G_t = \sum_{i=t}^{N-1} \gamma_{RL}^{i-t} r_i(X_i, A_i) + \gamma_{RL}^{N-t} g_N(X_N).$$

γ = 1: undiscounted return.

STATE VALUE: BEFORE CHOOSING AN ACTION

$$v_t^\pi(x) = E_\pi\left[G_t \mid X_t = x\right].$$

How good is this storage state when future actions follow policy π?

ACTION VALUE: AFTER FIXING AN ACTION

$$q_t^\pi(x, a) = E_\pi\left[G_t \mid X_t = x, A_t = a\right].$$

How good is charging now, followed by future actions from policy π?

Battery example: paying now can be worthwhile if stored energy produces a larger return later.

An action is judged by its immediate reward and the opportunities it leaves for later.

Bellman equations: reward plus future value

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EVALUATE A FIXED POLICY π

$$q_t^{\pi}(x,a) = r_t(x,a) + \gamma_{RL} \int_E v_{t+1}^{\pi}(x') Q_t(dx'|x,a),$$

$$v_t^{\pi}(x) = \int_{D_t(x)} q_t^{\pi}(x,a) \pi_t(da|x).$$

The policy averages action values; a deterministic policy selects one action.

SELECT THE BEST ACTION

$$q_t^{*}(x,a) = r_t(x,a) + \gamma_{RL} \int_E v_{t+1}^{*}(x') Q_t(dx'|x,a),$$

$$v_t^{*}(x) = \text{¿} a \in D_t(x) q_t^{*}(x,a).$$

Expectation evaluates a policy; optimality compares actions to find the best attainable value.

Temporal-difference learning trains the critic

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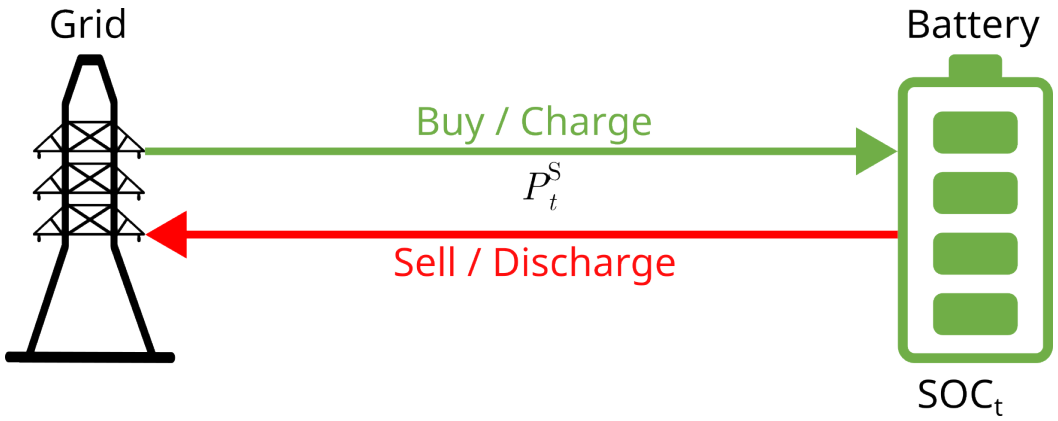
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Replay sample (x,a,r,x′,d) → Actor choose action → Critic estimate return → Actor update improve action

Policy gradients and actor–critic learning

Current state → Actor choose action → Critic estimate return → Actor update improve action

Battery system and operating task



- Positive power: buy and charge.
  Negative: sell and discharge.
- Capacity: 200 kWh.
- Power: 20 kW; efficiency: 0.9.
- Terminal SOC must be at least 0.50.
- The numerical study uses no buy–sell price spread.

ECONOMIC OBJECTIVE · EUR

$J = \Sigma (S \times P \times \Delta t / 1000)$

Lower J: more battery profit or less factory cost.

FEASIBILITY · OWN UNITS

Storage and power bounds; temperature coupling; terminal requirements.

TIME

Complete 24-hour rollout versus optimization solve. Training effort is separate.

Buy at low prices, sell at higher prices, and preserve terminal storage.

Battery MDP: observation and action

SOC + price + time 24-step profile → Policy u ∈ [−1, 1] → Power mapping + SOC limit → Battery SOC update

STATE TRANSITION

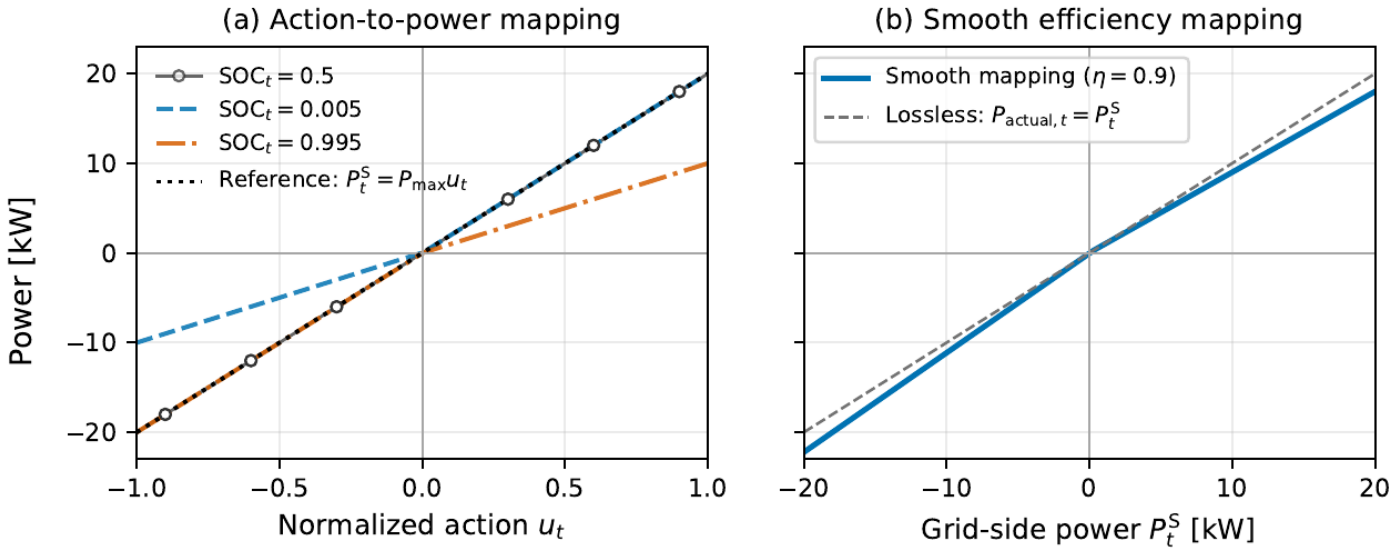
$$SOC_{t+1} = SOC_t + P_{actual,t} \frac{\Delta t}{E_{max}}$$

POWER CONVENTION

u > 0: charge; u < 0: discharge.
Grid and stored power differ because of efficiency.

One normalized command becomes grid-side and effective storage-side power.

Battery feasibility mapping and reward



- SOC gates attenuate power near empty/full storage.
- Clip storage power to available energy and remaining capacity.
- If clipped, recompute grid power before calculating cost.
- Reward = −grid cost − terminal SOC-shortfall penalty.

Additional variant: progressive reachability penalty before the final hour.

Power clipping enforces SOC bounds; terminal SOC remains a penalized requirement.

Battery economics: paired baseline comparisons

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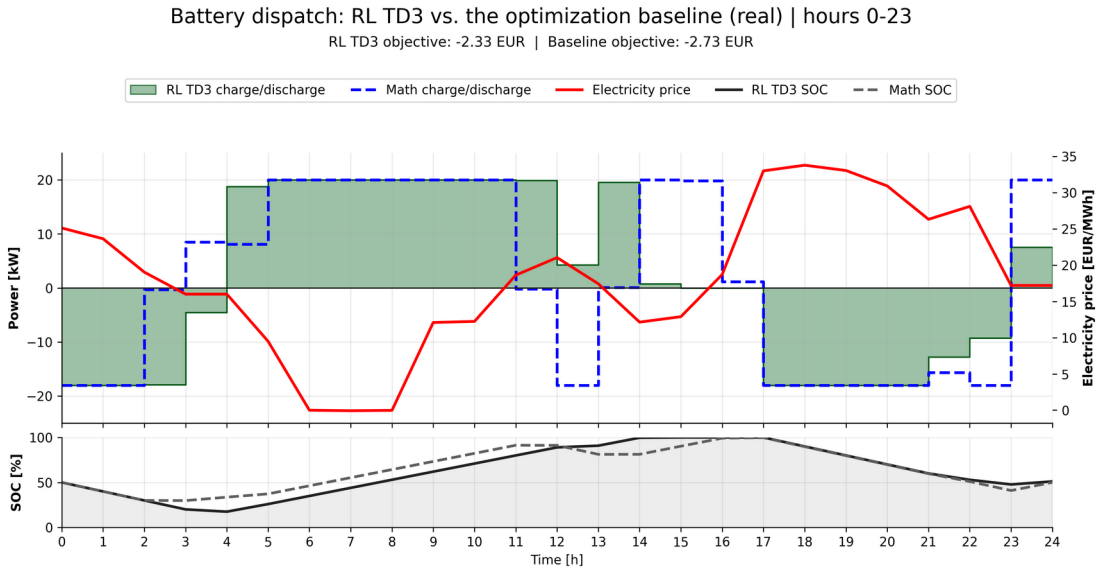
SEPARATE 1000-WINDOW EVALUATIONS · EUR

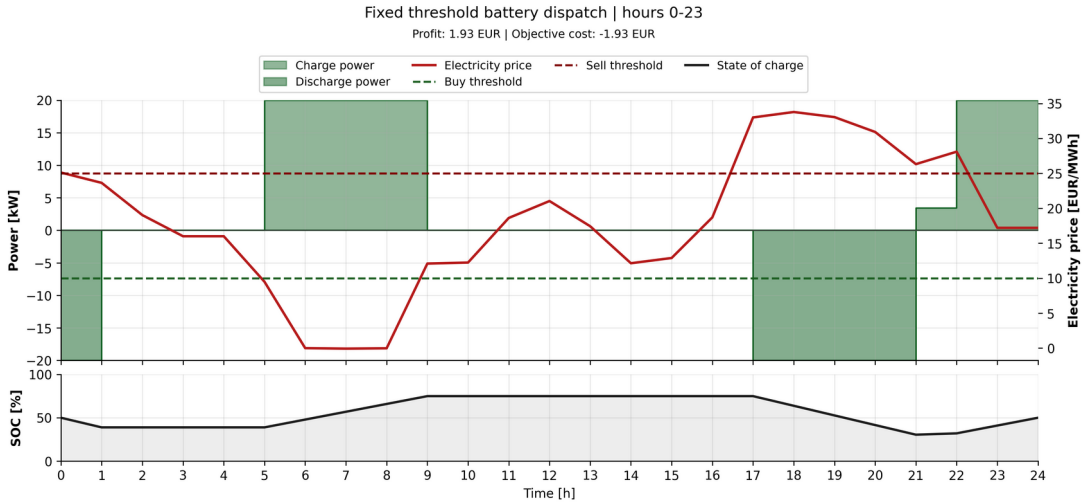
Metric	TD3 evaluation	SAC evaluation
Mean policy profit	3.0223	2.8911
Mean paired IPOPT profit	3.1851	3.1126
Mean signed objective gap	+0.1628	+0.2215
Gap standard deviation	0.1284	0.0811

Profit = −J. Each gap compares a policy with IPOPT on same windows.

Reported paired gaps: TD3 +0.1628 EUR; SAC +0.2215 EUR.

Battery dispatch on one matched price window





TD3 and IPOPT: forecast-based dispatch

Threshold: charge ≤ 10; discharge ≥ 25 EUR/MWh

≈ 20.4% higher TD3 profit than fixed thresholds, for this window only.

The threshold controller includes a terminal-SOC safeguard.

Battery runtime and training effort

Method	Mean runtime [s]
IPOPT	0.5605
TD3	0.0082
SAC	0.0126

Measured on AMD Ryzen 7 7840HS for complete 24-hour schedule.

68×

Reported TD3 speedup

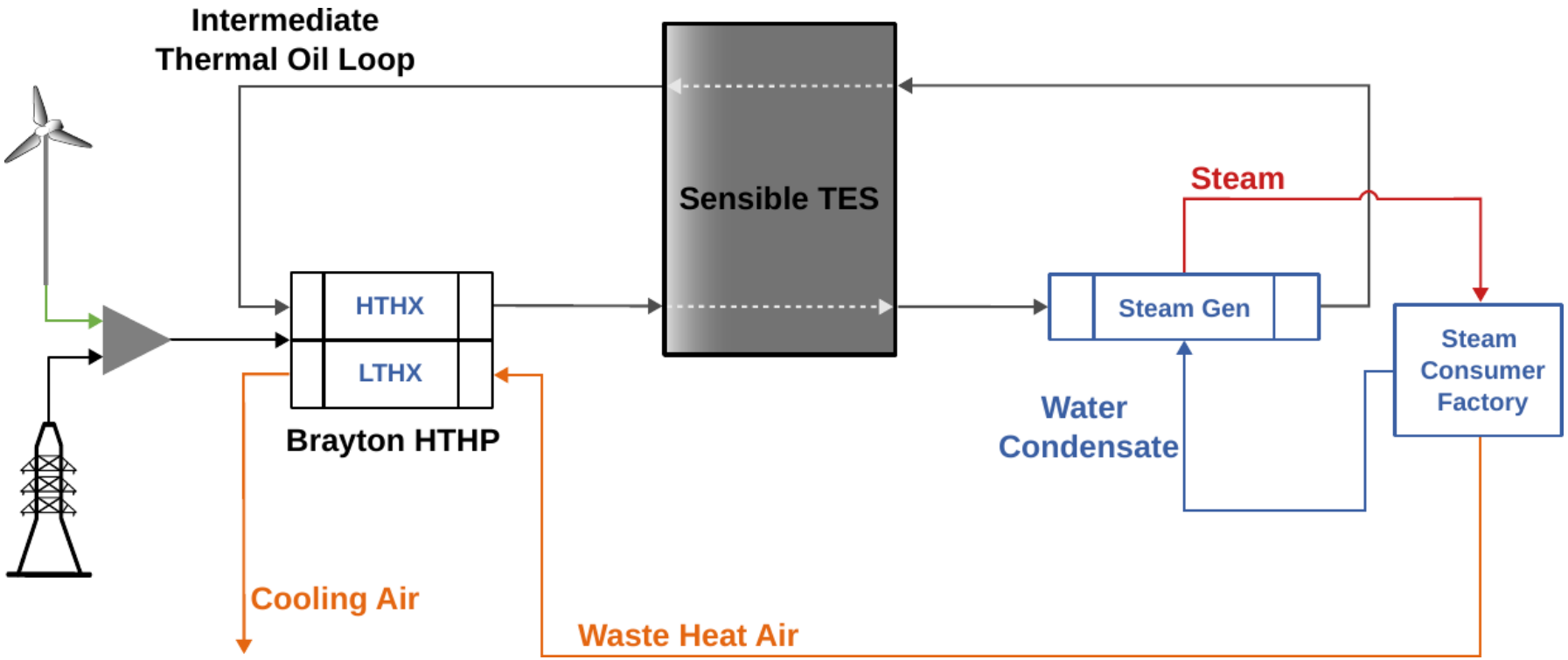
Online timing excludes training.

OFFLINE TRAINING

TD3: 23 minutes
SAC: 36 minutes

About 8 ms per complete TD3 rollout, after 23 minutes of offline training.

Cookie-factory energy utility system



Adapted from Kyriakidis et al. (2024), Fig. 2; original system: Walden et al. (2023).

Five controls: β_1, β_2, R, mass flow, HTHX inlet temperature.

Factory MDP and reduced action space

Observation: TES state, price, wind, time, terminal gap, previous controls/diagnostics, forecasts.

DISCHARGE

$b < -0.05$

$\beta_1 = 1$

IDLE

$|b| \leq 0.05$

$\beta_1 = \beta_2 = 1$

CHARGE

$b > 0.05$

$\beta_2 = 1$

HTF mass flow: 5–16 kg/s per HTHP; three units in parallel

Mode mapping bypasses the inactive path and prevents simultaneous TES charge/discharge.

Two policy outputs encode the storage mode and mass flow; other setpoints are derived.

Deriving compressor speed from coupling

Mode + mass flow + TES state → Required outlet temperature → Cubic surrogate solve for R

BOUNDED REAL ROOT EXISTS

$R \in [0.8, 1.53]$
Choose root nearest previous R.
Initial reference: 1.165.

NO BOUNDED ROOT

Apply bounded minimum-mismatch value.
Record the remaining F_4 residual.

A derived operating point can still be infeasible if no bounded root exists.

Factory reward and constraint handling

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HANDLED THROUGH ACTION STRUCTURE

Mode exclusivity
Bypass bounds
Mass-flow bounds

PENALIZED IN THE REWARD

State and inlet bounds; heat flow; F_4 coupling; grid power; terminal cycle.

Training bound	Value	Penalty weights
TES / inlet temperature	183–324 / 177–250 °C	State 20; inlet 10
Heat-flow / grid-power limit	5000 / 5000 kW	Heat 5000; grid 3000
Coupling / terminal cycle	Normalized residuals	F_4 500; cycle 3000

Reward = −grid cost − weighted normalized violations

Empirical normalized penalties encourage feasibility but do not guarantee it.

Factory economics over 200 matched windows

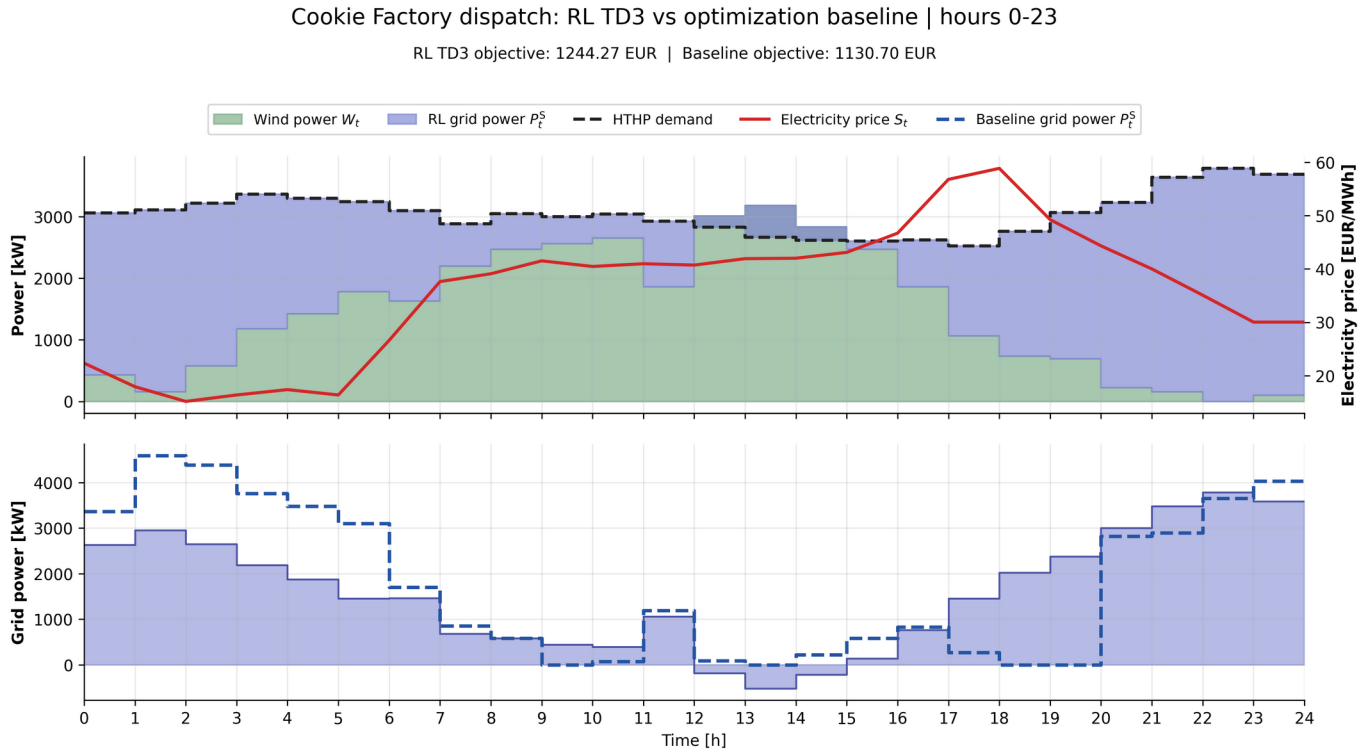
Metric	Math	TD3	SAC
Mean cost [EUR]	2012.2433	2081.3496	2092.5760
Mean signed gap [EUR]	—	69.1063	80.3327
Mean abs. relative gap	—	5.66%	6.31%

INTERPRETING THE GAPS

Schedules include violations. These are descriptive cost differences, not gaps between equally feasible schedules.

TD3 has the smaller RL objective gap; optimization has the lowest reported mean cost.

Factory dispatch: economics and feasibility



113.57

EUR example gap

About 10.0% relative to baseline.

TD3: 1244.27 EUR
IPOPT: 1130.70 EUR

Hours 12–15: TD3 grid power < 0 violates the lower bound.

IPOPT shifts grid purchases more strongly toward lower-price hours.

This example shows imperfect load shifting and a violation of the grid-power bound.

Factory feasibility: the main limitation

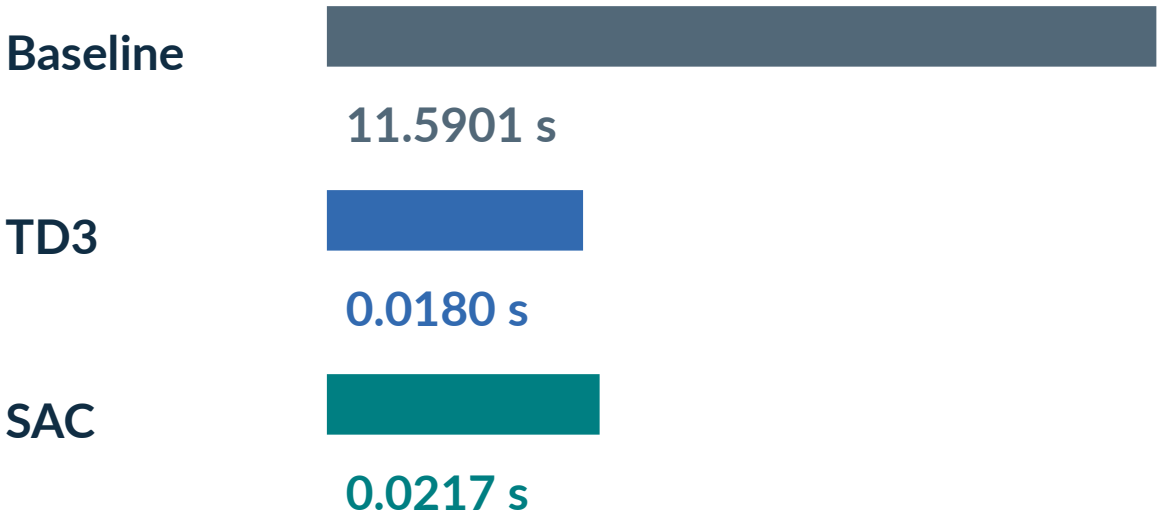
Constraint metric	Unit	SAC	TD3	Baseline
F_4 coupling residual	°C	0.4555	0.5300	0
Heat-flow-bound violation	kW	0.5989	3.5719	0
Grid-power violation	kW	98.3749	68.5787	0
Terminal cyclic deviation	°C	1.8380	2.1386	0

TD3: smaller grid-power violation. SAC: smaller F_4, heat-flow, and cyclic deviations.

First three rows: mean sums over 24 steps. Terminal row: mean absolute endpoint error.
Mean inlet/state-bound violations are zero for both policies.

Both RL policies have nonzero violations; the baseline reports zero.

Factory runtime and offline cost



in logarithmic scale (0.001–100 s).

Timing includes reset, predictions, transitions, cubic solves, and residual checks.
Model loading and training excluded; same hardware; SAC speedup 534×.

645×

Reported TD3 speedup

≈ 18 ms per 24-hour rollout.

OFFLINE TRAINING

TD3: 1.24 hours
SAC: 2.81 hours

What changes across the two cases?

TD3 evidence	Battery	Factory
Economic difference	0.16 EUR paired gap	69.10 EUR
Reported online speedup	68×	645×
Feasibility limitation	Small terminal shortfalls	Several process/grid violations

Scope and limitations

Fixed full-horizon forecasts · Simulator-based evaluation · No real-plant validation
Separate battery samples · Independent held-out performance is unestablished.
Models and economic scales differ; raw EUR gaps do not isolate complexity.

These case studies show a qualified trade-off between speed, economics, and feasibility.

Next steps toward reliable operation

RL suggestion → Feasibility restoration or IPOPT refinement → Applied setpoint

01 FEASIBILITY

Action projection; improve feasibility maps and derived R.

02 ROBUSTNESS

Rolling-horizon updates; forecast errors; disturbances and mismatch.

03 VALIDATION

RL warm starts; BO-IPOPT comparison; higher-fidelity or plant tests.

Proposed improvements target feasibility, forecast uncertainty, and plant realism.

Conclusion and questions

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RL learns economically meaningful continuous operating decisions.

TD3 combines small reported paired objective gaps with very low online computation time.

The optimization baseline retains the best reported mean objective and more reliable feasibility.

Thank you — Questions?

Rapid learned setpoints are promising; reliable constraint satisfaction remains essential.